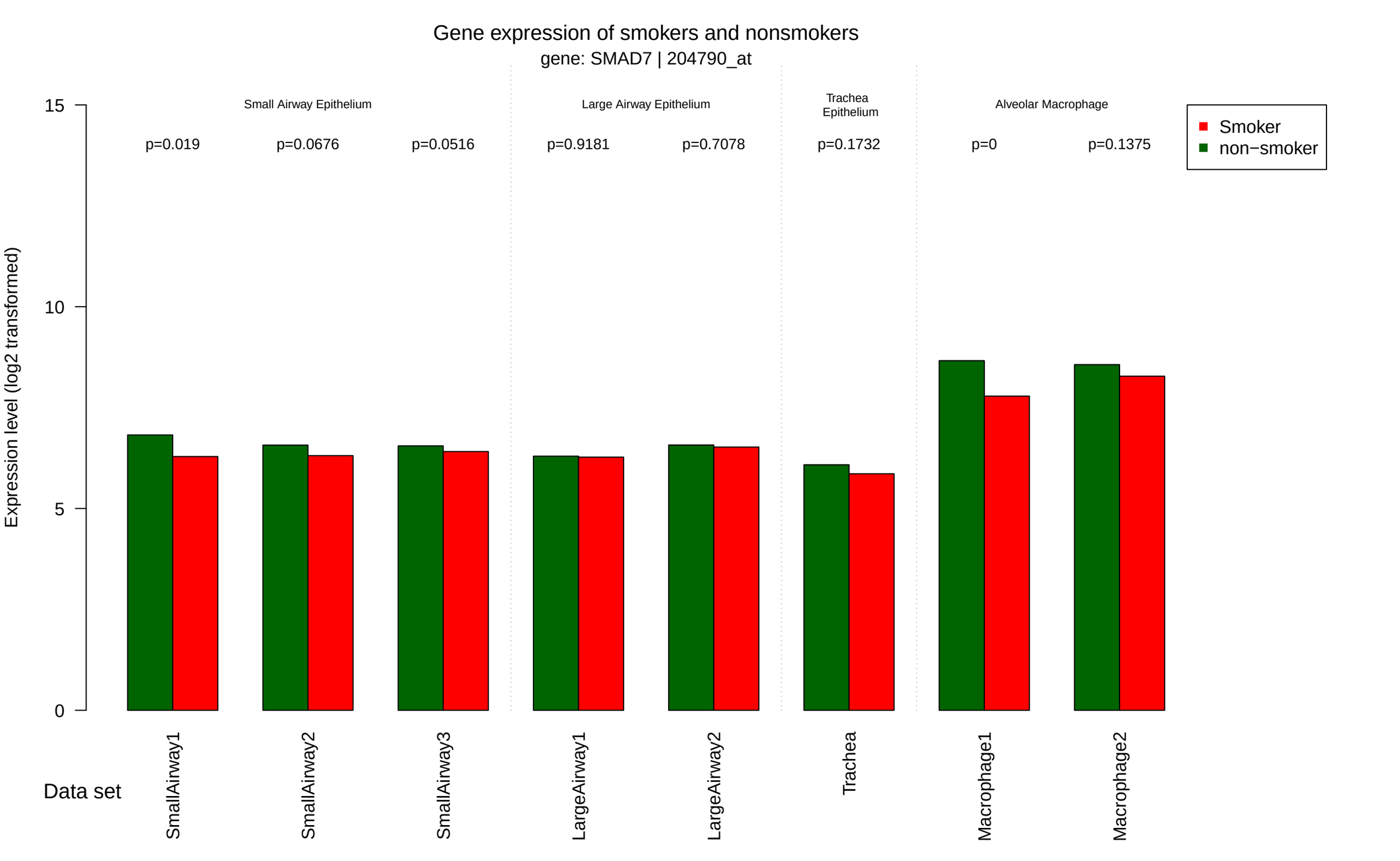
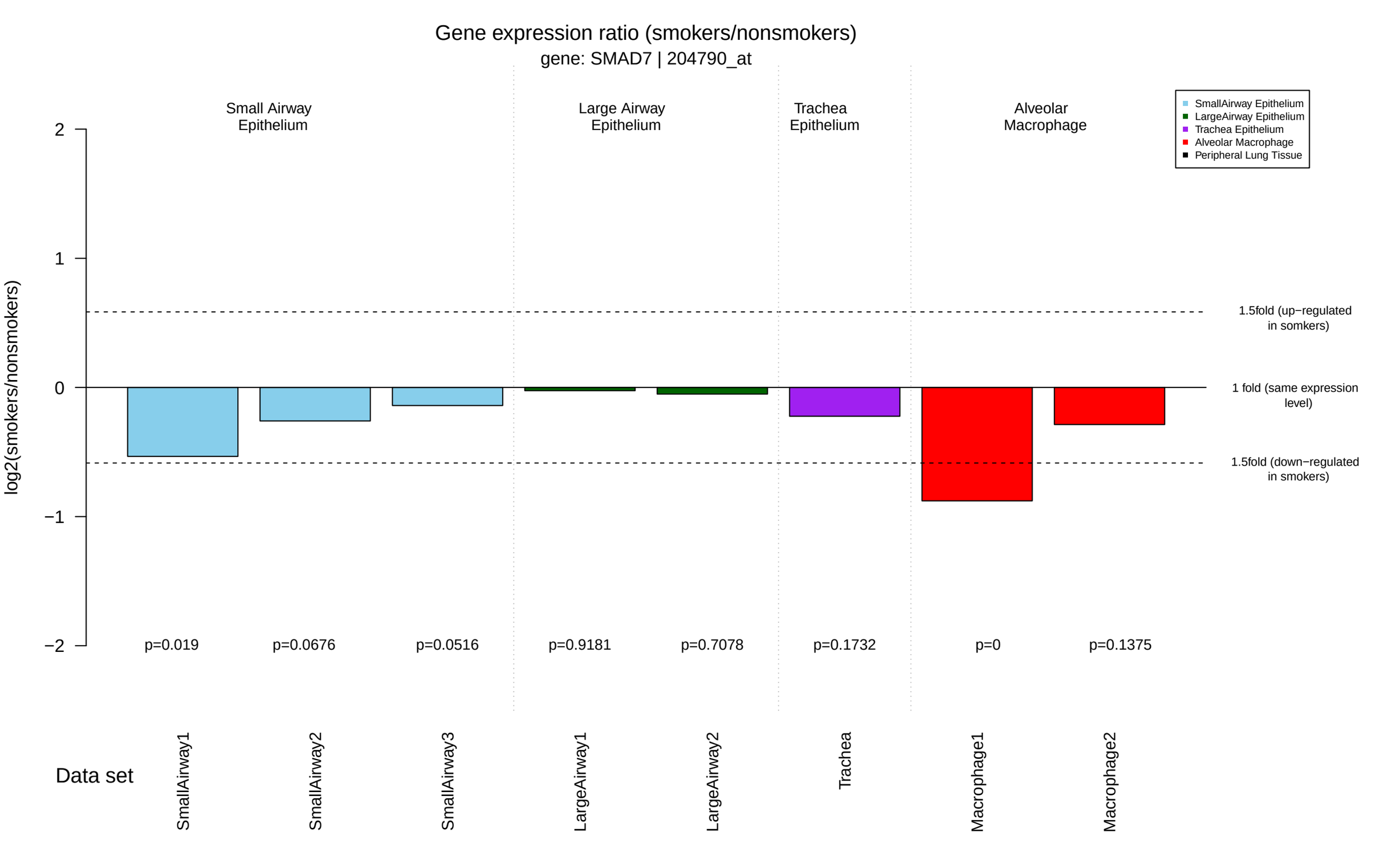
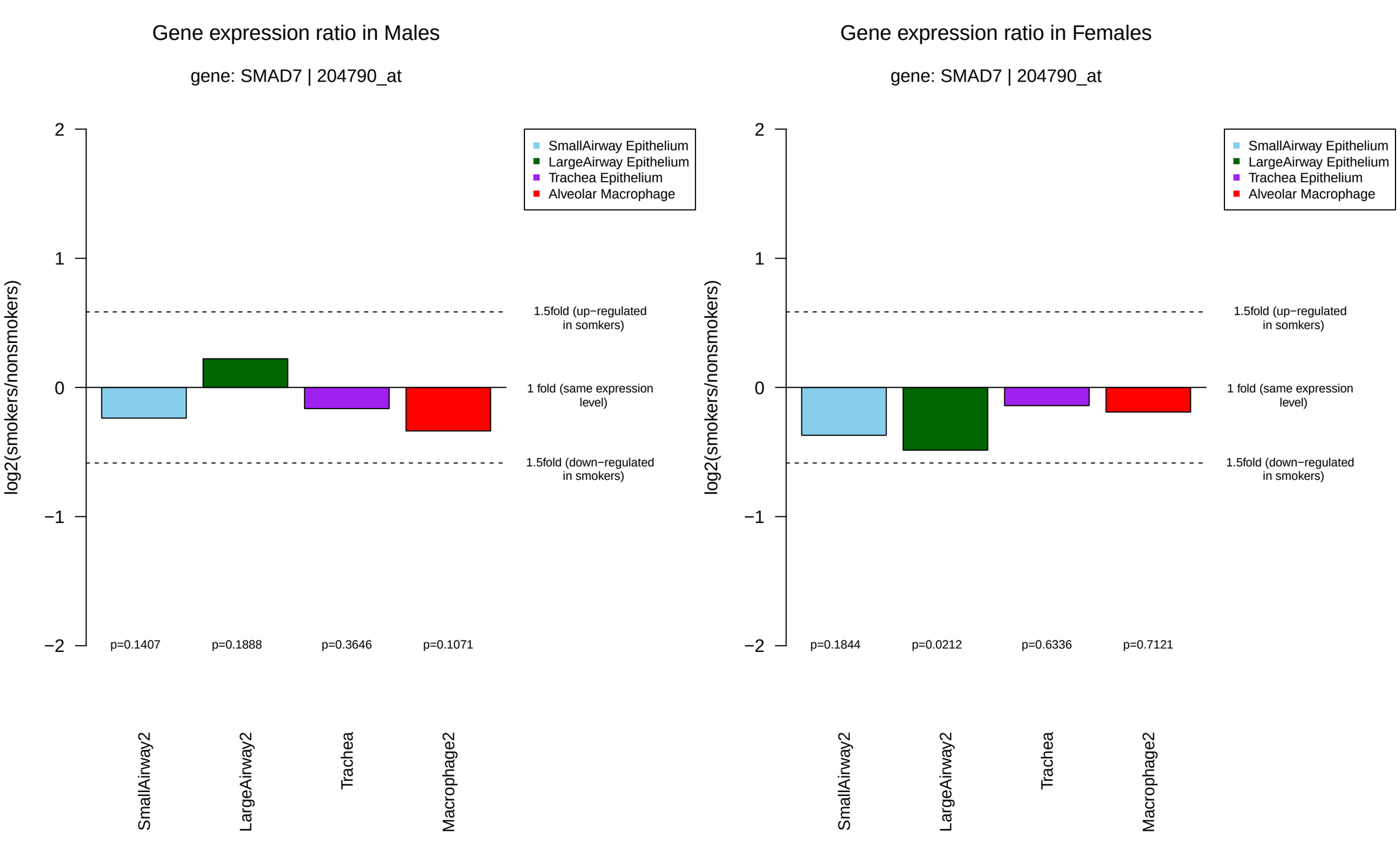
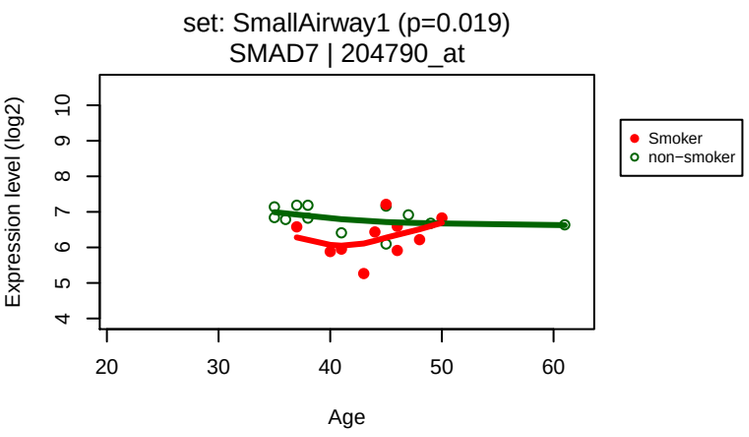
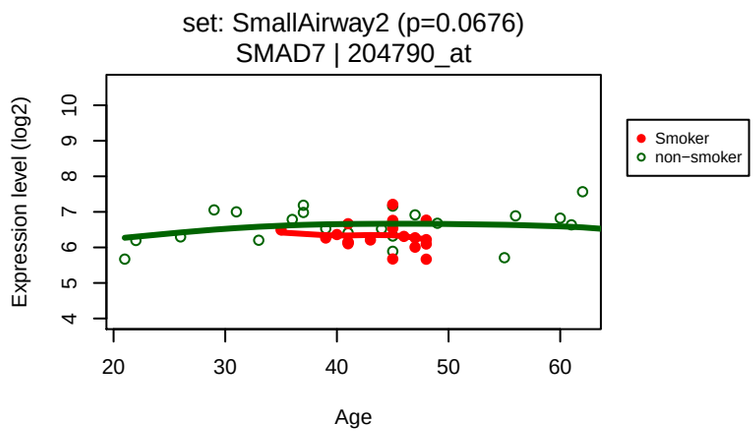
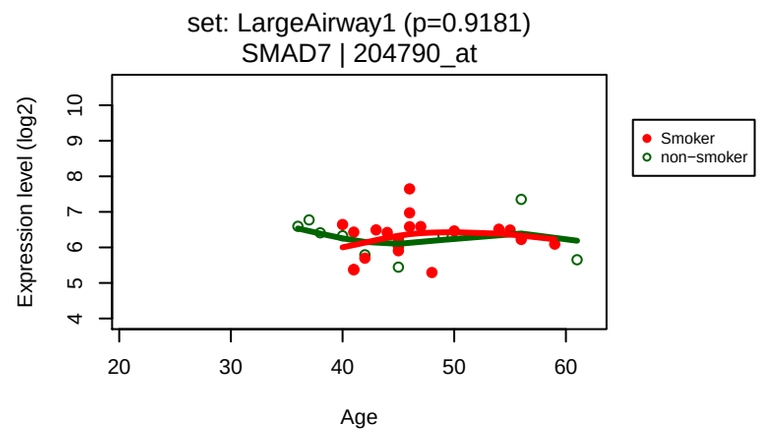

Gene expression of smokers and nonsmokers
gene: SMAD7 | 204790_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.019
p=0.0676
p=0.0516
p=0.9181
p=0.7078
p=0.1732
p=0
p=0.1375
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

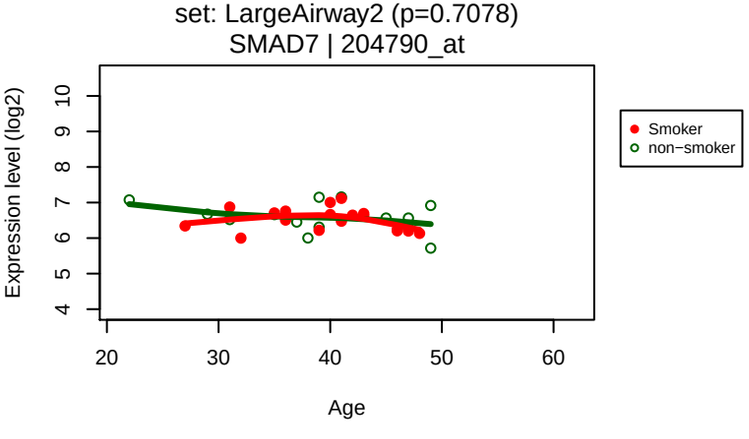

Gene expression ratio (smokers/nonsmokers)
gene: SMAD7 | 204790_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.019
p=0.0676
p=0.0516
p=0.9181
p=0.7078
p=0.1732
p=0
p=0.1375
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

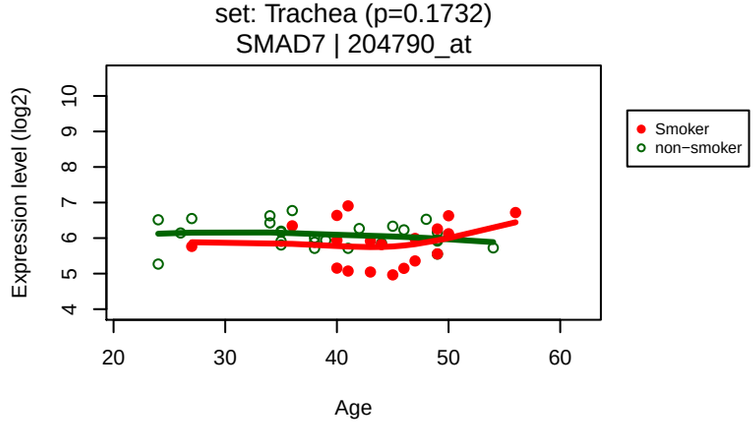

Gene expression ratio in Males
gene: SMAD7 | 204790_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1407
p=0.1888
p=0.3646
p=0.1071
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SMAD7 | 204790_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1844
p=0.0212
p=0.6336
p=0.7121
SmallAirway2
LargeAirway2
Trachea
Macrophage2

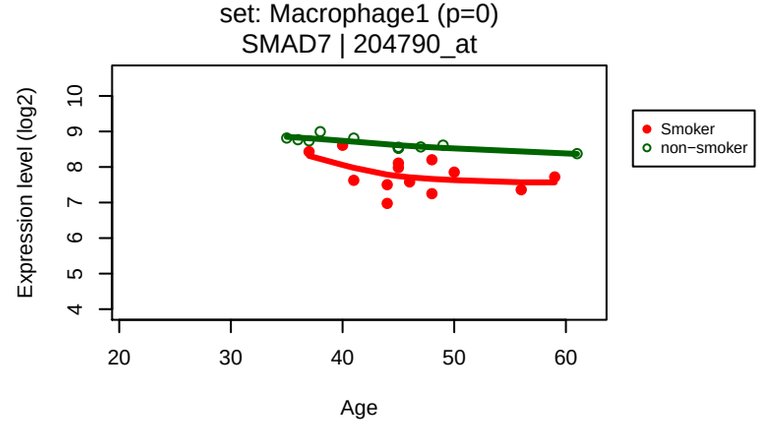

set: SmallAirway1 (p=0.019)
SMAD7 | 204790_at
Expression level (log2)
Age
Smoker
non−smoker

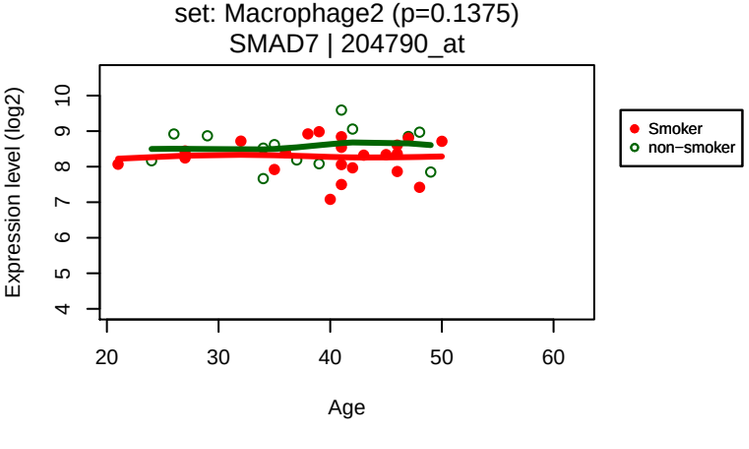

set: SmallAirway2 (p=0.0676)
SMAD7 | 204790_at
Expression level (log2)
Age
Smoker
non−smoker

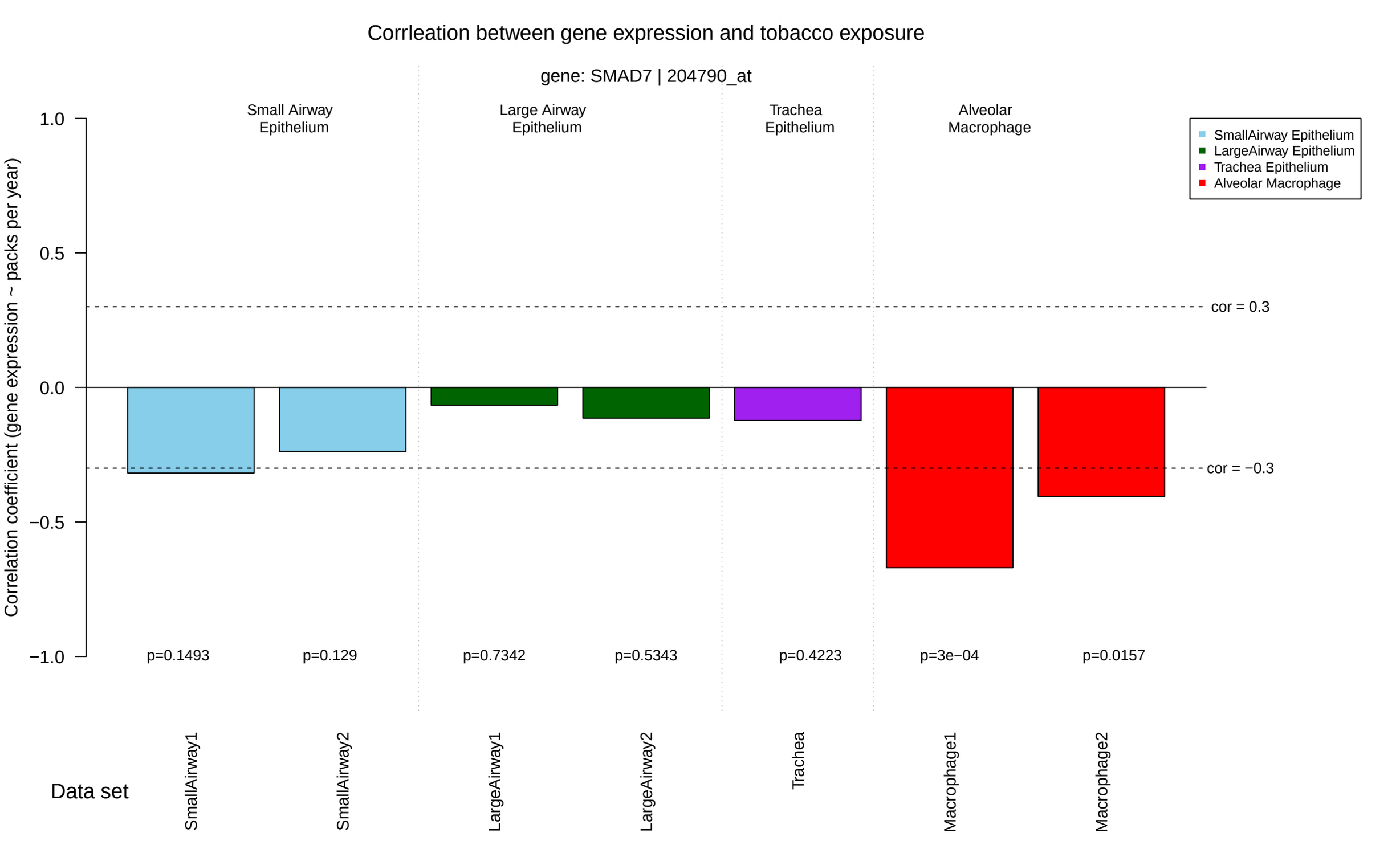

set: LargeAirway1 (p=0.9181)
SMAD7 | 204790_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7078)
SMAD7 | 204790_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1732)
SMAD7 | 204790_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0)
SMAD7 | 204790_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1375)
SMAD7 | 204790_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SMAD7 | 204790_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1493
p=0.129
p=0.7342
p=0.5343
p=0.4223
p=3e−04
p=0.0157
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2